

有界闭集上连续函数之性质:

1. 有界性.
2. 有最大最小值.
3. 有界闭集连通: 介值定理.

推论 2.15. (零点存在性定理). 设 $E \subset \mathbb{R}^2$ 一个区域.

$f \in C(E)$. $M_1(x_1, y_1) \in E$, $M_2(x_2, y_2) \in E$. $\Rightarrow$

$f(M_1) \cdot f(M_2) < 0$. 则 $\exists \bar{M}(\bar{x}, \bar{y}) \in E$. $\Rightarrow f(\bar{M}) = 0$.

证明: E 是区域 $\Rightarrow E$ 是连通集.

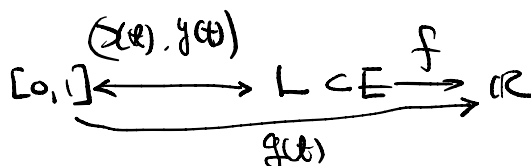
即 $\exists$ 折线 $L \subset E$, 连接 M_1, M_2 .

把 L 参数化: $L: \begin{cases} x=x(t) \\ y=y(t) \end{cases} t \in [0, 1]$

$x(t), y(t)$: 分段连续函数,

$(x(0), y(0)) \triangleq M_1(x_1, y_1)$

$(x(1), y(1)) = M_2(x_2, y_2)$.



$f(x, y)$ 在 E 中连续: $g(t) \triangleq \underline{f(x(t), y(t))} \in C[0, 1]$.

$g(0) = f(x(0), y(0)) = f(M_1)$
 $g(1) = f(x(1), y(1)) = f(M_2)$
 $\left. \begin{array}{l} g(0) = f(M_1) \\ g(1) = f(M_2) \end{array} \right\} g(0) \cdot g(1) < 0$

由介值定理: $\exists \bar{t} \in (0, 1) \Rightarrow g(\bar{t}) = 0$.

$(x(\bar{t}), y(\bar{t})) = (\bar{x}, \bar{y}) \in L \subset E$. $f(\bar{x}, \bar{y}) = g(\bar{t}) = 0$. #

定理 2.15. (一致连续性). 有界闭集上的连续函数是一致连续的. 即 设 $E \subset \mathbb{R}^2$ 是有界闭集. $f \in C(E)$, 则

f 在 E 上是 ϵ -一致连续. 即: $\forall \epsilon > 0, \exists \delta > 0, \exists$

$\forall M_1(x_1, y_1), M_2(x_2, y_2) \in E, r(M_1, M_2) < \delta$. 有

$$|f(M_1) - f(M_2)| < \epsilon.$$

证明: 用反证法. 假如 $f(x, y)$ 在 E 上不是 ϵ -一致连续;

即 $\exists \epsilon_0 > 0 \forall \delta > 0, \exists M_1, M_2 \in E$. 尽管 $r(M_1, M_2) < \delta$

但有 $|f(M_1) - f(M_2)| \geq \epsilon_0 > 0$.

$\delta = \frac{1}{k} (\xrightarrow{k \rightarrow \infty} 0)$. $\exists M_{1k}, M_{2k} \in E, r(M_{1k}, M_{2k}) < \frac{1}{k}$.

但仍有 $|f(M_{1k}) - f(M_{2k})| \geq \epsilon_0 > 0$.

$\because E$ 是有界闭集. $\{M_{1k}\} \subset E$ 有界点列.

由聚点定理: $\exists M_{1k_j'} \rightarrow P_0$,

$\{M_{2k_j'}\} \subset E$: 有界. $\exists$ 子列 $\exists M_{2k_j''} \rightarrow Q_0$.

$\Rightarrow M_{1k_j'} \rightarrow P_0$

$M_{2k_j''} \rightarrow Q_0$

$$r(M_{1k_j'}, M_{2k_j''}) < \frac{1}{k_j'} \rightarrow 0:$$

$$\downarrow \quad \downarrow$$
$$r(P_0, Q_0) = 0: \quad P_0 = Q_0;$$

由于 $|f(M_{1k_j'}) - f(M_{2k_j''})| \geq \epsilon_0 > 0$.

$k_j' \rightarrow \infty$: $f(M)$ 连续: $0 = |f(P_0) - f(P_0)| \geq \epsilon_0 > 0$

矛盾! $\therefore f(x, y)$ 在 E 上 ϵ -一致连续 #.

第14章. 偏导数和全微分.

§1. 偏导数和全微分的概念.

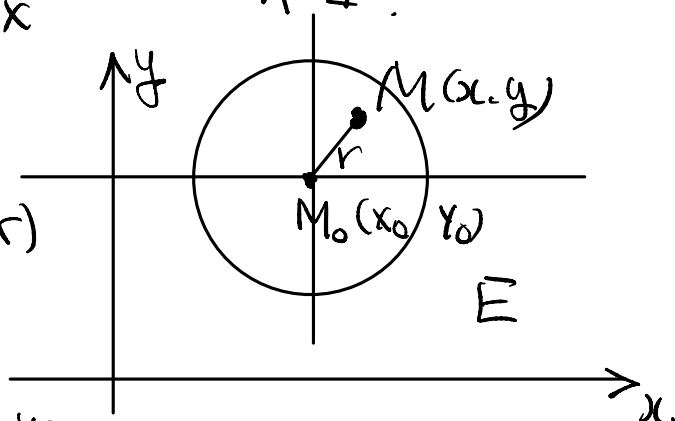
Recall: 一元函数 $y=f(x)$, $x \in O(x_0, r)$.

$$f'(x_0) := \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = A \exists.$$



$$u = f(x, y): (x, y) \in O(M_0, r)$$

$\Delta x, \Delta y$



$$\Delta u = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0).$$

$$\frac{\Delta u}{\Delta x}, \frac{\Delta u}{\Delta y}, \quad \frac{\Delta u}{\rho} ?$$

$$\begin{aligned} M_0 &\rightarrow M: \\ (x_0, y_0) &\rightarrow (x, y) \\ x - x_0 &= \Delta x \\ y - y_0 &= \Delta y \quad \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2} \end{aligned}$$

一. 偏导数的定义.

定义1.1. 设函数 $u = f(x, y)$ 在 $M_0(x_0, y_0)$ 的某个邻域中有定义. 对 x 的增量 Δx :

$$\Delta_x u = f(x_0 + \Delta x, y_0) - f(x_0, y_0)$$

$$\text{若 } \lim_{\Delta x \rightarrow 0} \frac{\Delta_x u}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = A \exists$$

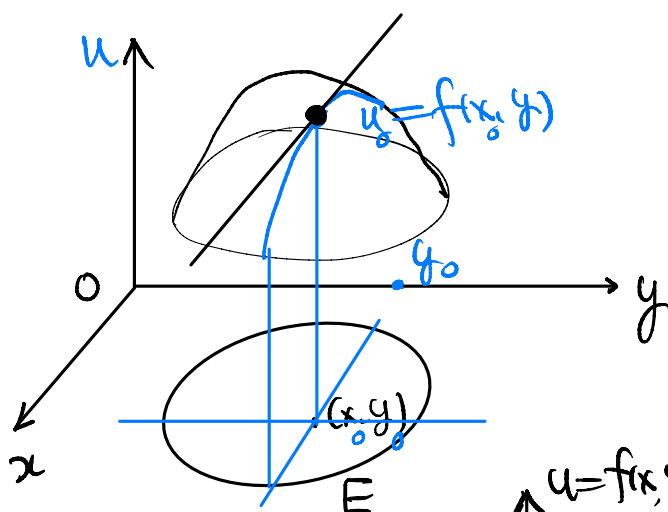
则称 A 是 $f(x, y)$ 在 $M_0(x_0, y_0)$ 关于 x 的偏导数

记为 $\frac{\partial u}{\partial x} \Big|_{(x_0, y_0)} = A$; $\frac{\partial f}{\partial x} \Big|_{(x_0, y_0)} \stackrel{\text{或}}{=} u_x(x_0, y_0) = f_x(x_0, y_0)$

同理: $\frac{\partial u}{\partial y} \Big|_{(x_0, y_0)} = \frac{\partial f}{\partial y} \Big|_{(x_0, y_0)} = u_y(x_0, y_0) = f_y(x_0, y_0)$

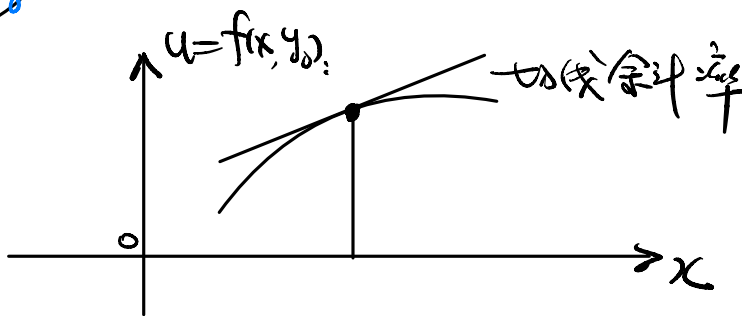
$$= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}$$

称 $f(x, y)$ 在 $M_0(x_0, y_0)$ 关于 y 偏导数
 (Partial derivative)



$u = f(x, y_0)$:

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = A$$



$A = \frac{\partial u}{\partial x} \Big|_{(x_0, y_0)}$:

曲面上二曲线: $\begin{cases} u = f(x, y) \\ y = y_0 \end{cases}$ 切线斜率.
 (The diagram shows arrows pointing from the equations to the corresponding parts of the 3D surface and the 2D graph.)

一般地: $n \in \mathbb{N}$ $u = f(x)$, $x \in O(x_0, r)$.

$x = (x_1, x_2, \dots, x_n)$: $x_0 = (x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})$

$1 \leq i \leq n$:

$$\Delta u = f(\underbrace{x_1^{(0)}, x_2^{(0)}, \dots, x_i^{(0)} + \Delta x_i, x_{i+1}^{(0)}, \dots, x_n^{(0)}}) - f(x_0)$$

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, \dots, 0), \quad e_i = (0, \dots, \overset{\text{ith}}{1}, \dots, 0)$$

$$e_n = (0, 0, \dots, 1)$$

$$\begin{aligned} \underbrace{x_0 + \Delta x_i e_i} &= \underbrace{(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})} + (0, \dots, \overset{\text{ith}}{\Delta x_i}, \dots, 0) \\ &= (x_1^{(0)}, x_2^{(0)}, \dots, \underbrace{x_i^{(0)} + \Delta x_i}, \dots, x_n^{(0)}) \end{aligned}$$

$$\Delta u = f(\underbrace{x_0 + \Delta x_i e_i}) - f(x_0)$$

若 $\lim_{\Delta x_i \rightarrow 0} \frac{\Delta u}{\Delta x_i} = A_i$ 存在, 则 A_i 就称为 $u=f(x)$

在 x_0 关于 x_i 的偏导数.

$$\frac{\partial u}{\partial x_i} \Big|_{x_0} = \frac{\partial f}{\partial x_i} \Big|_{x_0} = u_{x_i}(x_0) = f_{x_i}(x_0).$$

$$\triangleq \frac{d}{dx_i} f(x_1^{(0)}, x_2^{(0)}, \dots, \underbrace{x_i}, \dots, x_n^{(0)}) \Big|_{x_i=x_i^{(0)}}$$

例: $u = f(x, y, z) = \frac{xy^2}{z}$.

$$\frac{\partial u}{\partial x} = \frac{y^2}{z}; \quad \frac{\partial u}{\partial y} = 2 \frac{xy}{z}; \quad \frac{\partial u}{\partial z} = -\frac{xy^2}{z^2}$$

例 $f(x) = |x| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$, $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

$$\begin{aligned}
\frac{\partial f}{\partial x_i} &= \frac{d}{dx_i} (x_1^2 + x_2^2 + \dots + x_i^2 + \dots + x_n^2)^{\frac{1}{2}} \\
&= \cancel{\frac{1}{2}} (x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}-1} \cdot \cancel{2} x_i \\
&= (x_i^2)^{-\frac{1}{2}} \cdot x_i \\
&= |x|^{-1} \cdot x_i = \frac{x_i}{|x|}.
\end{aligned}$$

$\therefore |x|^2 = x_1^2 + x_2^2 + \dots + \underbrace{x_i^2}_{\text{circled}} + \dots + x_n^2,$

$$\cancel{2}|x| \cdot \frac{d|x|}{dx_i} = \cancel{2} x_i$$

$$\frac{\partial f}{\partial x_i} = \frac{d|x|}{dx_i} = \frac{x_i}{|x|}$$

Ex. $u = f(x, y, z) = \ln \sqrt{x^2 + y^2 + z^2}$

$$= \frac{1}{2} \ln(x^2 + y^2 + z^2)$$

$$\frac{\partial u}{\partial x} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2 + z^2} \cdot 2x = \frac{x}{x^2 + y^2 + z^2},$$

$$\frac{\partial u}{\partial y} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2 + z^2} \cdot 2y = \frac{y}{x^2 + y^2 + z^2},$$

$$\frac{\partial u}{\partial z} = \frac{z}{x^2 + y^2 + z^2},$$

Ex. $u = f(x) = g(|x|) \quad |x| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

$$\frac{\partial u}{\partial x_i} = f'(\underline{x}) \cdot \frac{\partial \underline{x}}{\partial x_i} = f'(\underline{x}) \cdot \frac{x_i}{|\underline{x}|}.$$

二. 全微分

Recall: $y=f(x)$, 在 x_0 处可微:

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = A \Delta x + \underbrace{o(\Delta x)}_{\substack{\uparrow \\ \text{比 } \Delta x \text{ 更快地趋于 } 0}}.$$

$$\frac{o(\Delta x)}{\Delta x} \rightarrow 0 \quad (\Delta x \rightarrow 0).$$

$$A \Delta x: \text{主部}. \quad dy = A \Delta x = A dx.$$

$$dy = A dx; \quad A = f'(x_0) = \frac{dy}{dx};$$

定义 1.2. 设 $u=f(x,y)$ 定义于 (x_0, y_0) 邻域 $O((x_0, y_0), r)$

对于 $u=f(x,y)$, 在 (x_0, y_0) 邻域中: $(\Delta x, \Delta y)$

$$\text{若 } \Delta u = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \quad (x, y) = (x_0 + \Delta x, y_0 + \Delta y)$$

$$= A \Delta x + B \Delta y + \underbrace{o(\rho)}_{\substack{\uparrow \\ \text{比 } \rho \text{ 更快地趋于 } 0}}; \quad \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2} \\ = r(x, y, (x_0, y_0))$$

$$\frac{o(\rho)}{\rho} \rightarrow 0 \quad (\rho \rightarrow 0) \quad \underline{A, B \in \mathbb{R}}$$

则称 $u=f(x,y)$ 在 (x_0, y_0) 可微 或

$A \Delta x + B \Delta y$ 为 $f(x,y)$ 在 (x_0, y_0) 处全微分.

$$\text{记 } du = A \Delta x + B \Delta y = df|_{(x_0, y_0)} = df(x_0, y_0).$$

注: ① 若 $u=f(x,y)$ 在 (x_0, y_0) 可微:

$$\begin{aligned}\Delta u &= f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \\ &= A\Delta x + B\Delta y + o(\rho) \quad \xrightarrow[\substack{\rho \rightarrow 0 \\ (x,y) \rightarrow (x_0,y_0)}}{0}\end{aligned}$$

$$\therefore \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = f(x_0, y_0)$$

$\Rightarrow f(x,y)$ 在 (x_0, y_0) 连续.

可微 $\Rightarrow$ 连续

② 若 $u=f(x,y)$ 在 (x_0, y_0) 可微: $\frac{\Delta y}{\Delta x} = 0$:

$$\Delta u = f(x_0 + \Delta x, y_0) - f(x_0, y_0)$$

$$= A\Delta x + o(\rho)$$

$$\Rightarrow f(x, y_0) \text{ 在 } x_0 \text{ 可微. 且 } A = \left. \frac{\partial f}{\partial x}(x, y_0) \right|_{x_0}$$

$$\text{同理: } \Delta x = 0: \Rightarrow B = \left. \frac{\partial f}{\partial y}(x_0, y) \right|_{y_0}$$

可微 $\Rightarrow$ 偏导数存在

$$\Downarrow$$
$$\text{且 } du = A\Delta x + B\Delta y = \left. \frac{\partial f}{\partial x} \right|_{(x_0,y_0)} \Delta x + \left. \frac{\partial f}{\partial y} \right|_{(x_0,y_0)} \Delta y$$

$$\text{若 } u = f(x, y) = x: \quad du = dx = \frac{\partial f}{\partial x} \Delta x = \Delta x;$$

$$\text{若 } u = f(x, y) = y: \quad du = dy = \frac{\partial f}{\partial y} \Delta y = \Delta y.$$

$$dx = \Delta x, \quad dy = \Delta y \Rightarrow$$

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

$$\text{一般地: } u = f(x) \quad x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

$$\Delta x = (\Delta x_1, \Delta x_2, \dots, \Delta x_n),$$

$$\exists A_i \in \mathbb{R}.$$

$$\rho = r(x, x + \Delta x) = |\Delta x| = \sqrt{\sum_{i=1}^n (\Delta x_i)^2}$$

$$\Delta u = f(x + \Delta x) - f(x)$$

$$= f(x_1 + \Delta x_1, x_2 + \Delta x_2, \dots, x_n + \Delta x_n) - f(x)$$

$$= A_1 \cdot \Delta x_1 + A_2 \cdot \Delta x_2 + \dots + A_n \cdot \Delta x_n + o(\rho)$$

高阶无穷小

$$\frac{o(\rho)}{\rho} \rightarrow 0 \quad (\rho \rightarrow 0)$$

$$\text{若 } u = f(x) \text{ 可微, } \sum_{i=1}^n A_i \Delta x_i: f(x) \text{ 可微.}$$

$$du = \sum_{i=1}^n A_i \Delta x_i = \sum_{i=1}^n A_i dx_i, \quad \Delta x_i = dx_i$$

$$\underline{A} \quad A_i = \frac{\partial f}{\partial x_i}$$

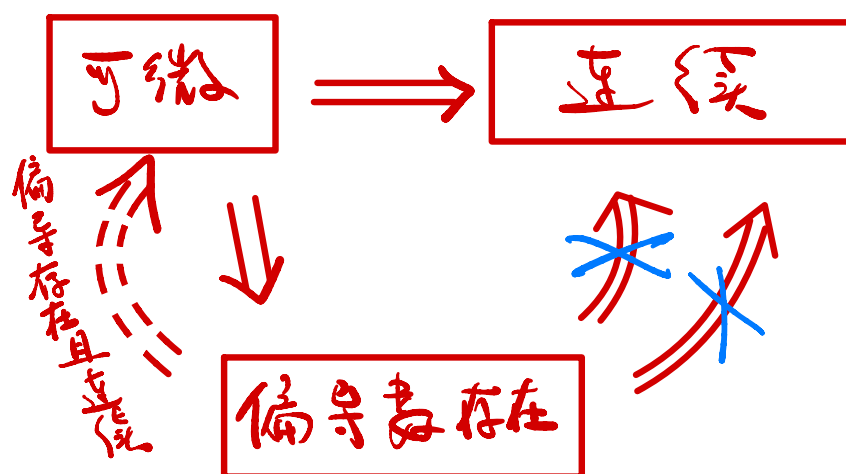
$$\text{例 } u = f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f(x, 0) = \begin{cases} 0, & x \neq 0 \\ 0, & x = 0 \end{cases} \Rightarrow f(x, 0) \equiv 0,$$

$$\therefore \frac{\partial f}{\partial x}(0, 0) = 0;$$

$$\text{同理: } \frac{\partial f}{\partial y}(0, 0) = 0,$$

但: $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \nexists$ 不连续



定理 1.3. $u = f(x, y)$ 在 $O(M_0, r)$ 中每点有偏导数.

$f_x(x, y), f_y(x, y) \exists$ 且连续,

$\Rightarrow u = f(x, y)$ 在 M_0 可微